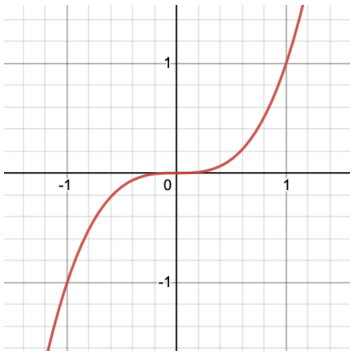
1 Differentiability of Inverse Functions

Suppose f is differentiable and one-to-one on an interval. Since f is differentiable, its graph has no gaps or breaks, no corners or kinks, and no vertical tangent lines.



1. If the graph of f has no gaps or breaks, no corners or kinks, and no vertical tangent lines, could the graph of f^{-1} have one of these? Sketch the graph of the inverse of the function above. What do you conclude about differentiability of f^{-1}? Explain.

While reflecting the graph of a 1-1 differentiable function $y = f(x)$ across the line $y = x$ cannot produce any discontinuity or sharp point, if the graph of f has a horizontal tangent line at x_0, then reflection of the graph across $y = x$ will map this to a vertical tangent line. Therefore, this is the only way in which f^{-1} can fail to be differentiable.

Theorem. Suppose f is differentiable and one-to-one on an interval. If $f'(x) \neq 0$, then f^{-1} is differentiable at $y = f(x)$ and

$$(f^{-1})'(y) = \frac{1}{f'(x)}, \qquad \text{where } x = f^{-1}(y). \tag{1}$$

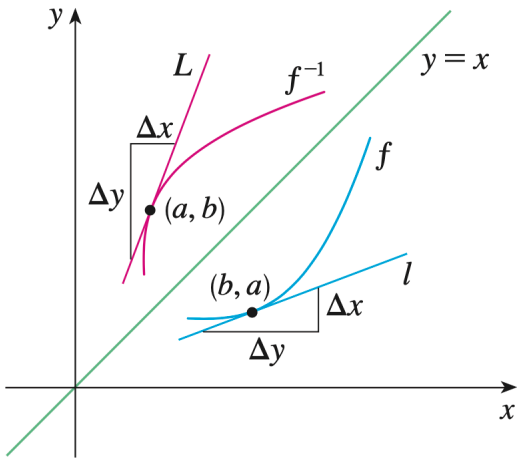
The proof below uses the fact that if f is continuous and 1-1 on an interval (a,b), then f^{-1} is continuous on the interval $f((a,b))$. (We proved this in handout 3.2 when we talked about continuity.)

Proof. Let x_0 be a point where f is differentiable. Then f is continuous at x_0, and therefore f^{-1} is continuous at $y_0 = f(x_0)$, so we have $\lim_{y\to y_0} f^{-1}(y) = f^{-1}(y_0)$. Since $x = f^{-1}(y)$, we can also write this as $\lim_{y\to y_0} x = x_0$. It follows that, if $f'(x_0) \neq 0$, then at $y_0 = f(x_0)$ we have

$$\begin{aligned}
(f^{-1})'(y_0) &= \lim_{y\to y_0} \frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} \\
&= \lim_{y\to y_0} \frac{x - x_0}{f(x) - f(x_0)} \\
&= \lim_{x\to x_0} \frac{x - x_0}{f(x) - f(x_0)} \qquad (\text{since } x \to x_0 \text{ as } y \to y_0) \\
&= \lim_{x\to x_0} \frac{1}{\frac{f(x)-f(x_0)}{x-x_0}} \\
&= \frac{\lim_{x\to x_0} 1}{\lim_{x\to x_0} \frac{f(x)-f(x_0)}{x-x_0}} \\
&= \frac{1}{f'(x_0)}.
\end{aligned}$$

□

We can understand the formula in Equation (1) geometrically, as follows. The figure below shows the graph of a differentiable function and the graph of its inverse.



The derivative $(f')^{-1}(a)$ is the slope of the tangent line to the graph of f^{-1} at (a,b). Since reflecting across the line $y=x$ has the effect of interchanging the x- and y-coordinates, we see that

$$(f^{-1})'(a) = \frac{\Delta y}{\Delta x} = \frac{1}{\Delta x/\Delta y} = \frac{1}{f'(b)}.$$

Now we turn to some examples.

2. Consider the function $f(x) = mx + b$.

   (a) Find $f'(x)$.

   (b) Find $f^{-1}(y)$, where $y = f(x)$.

   (c) Find $(f^{-1})'(y)$.

   (d) Verify that Equation (1) holds.

Remarks.

(a) Note that the prime on the left-hand-side of Equation (1) denotes differentiation with respect to y, while the prime on the right-hand-side denotes differentiation with respect to x. In Leibniz notation, this is

$$\frac{df^{-1}}{dy}(y_0) = \frac{1}{\frac{df}{dx}(x_0)}, \quad y_0 = f(x_0).$$

(b) Equation (1) tells us that we do not need to explicitly solve for f^{-1} in order to find (f^{-1})'. This is great news, since finding explicitly finding the inverse of a function can be very hard or impossible!

(c) There is a Khan Academy video on this in which he uses some notation which students in the past found very confusing. I would avoid his videos on this topic.

Example. The function $f(x) = \sqrt{x} + x^2 + 1$ is differentiable and one-to-one for $x \geq 0$. Note that $f(1) = 3$. Since $f'(x) = \frac{1}{2\sqrt{x}} + 2x$ and therefore $f'(1) = \frac{1}{2} + 2 = \frac{5}{2}$, we have $(f^{-1})'(3) = \frac{1}{f'(1)} = \frac{1}{\frac{5}{2}} = \boxed{\frac{2}{5}}$.

3. For the same function above, find $(f^{-1})'(19)$. (Note that $f(4) = 19$.)

Example. Use the values of the differentiable bijection f in the table below to compute the indicated derivatives or state the derivative cannot be determined.

x	-1	0	1	2	3
$f(x)$	2	3	5	6	7
$f'(x)$	$1/2$	2	$3/2$	1	$2/3$

a. $(f^{-1})'(5)$ **b.** $(f^{-1})'(2)$ **c.** $(f^{-1})'(1)$

For example, (a) $(f^{-1})'(5) = \frac{1}{f'(1)} = \frac{1}{\frac{3}{2}} = \frac{2}{3}$.

4. Solve (b) and (c).

5. Prove using our theorem that $\ln(x)$ is differentiable for $x > 1$ and is given by $\frac{d}{dx}\ln(x) = \frac{1}{x}$.